

Math Primer

(19. Named Distributions)

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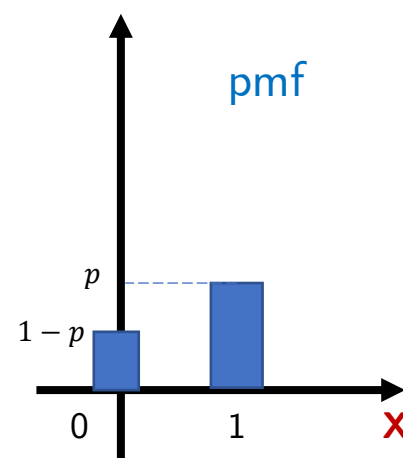
C.F. Gauss
(1777-1855)

(discrete distributions)

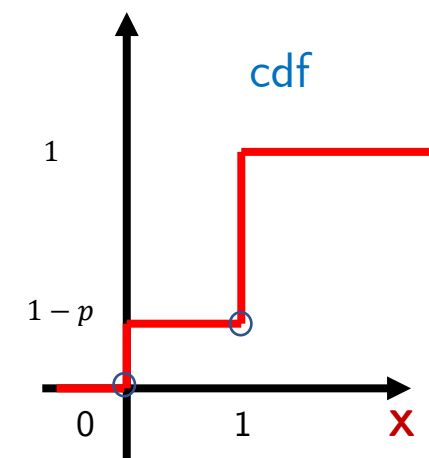
Bernoulli Distribution

- Used to model a phenomenon with binary outcomes (dichotomous) : [True/False](#), [Success/Failure](#), [Default/No Default](#) etc.

Parameters	$p \in (0,1)$
Support	$x \in \{0, 1\}$
PMF	$\begin{cases} 1 - p, & x = 0 \\ p, & x = 1 \end{cases}$
PMF (alternate)	$p^x(1 - p)^{1-x}$
Mean	p
Variance	$p(1 - p)$



$$P_X(x) = p^x(1 - p)^{1-x}$$



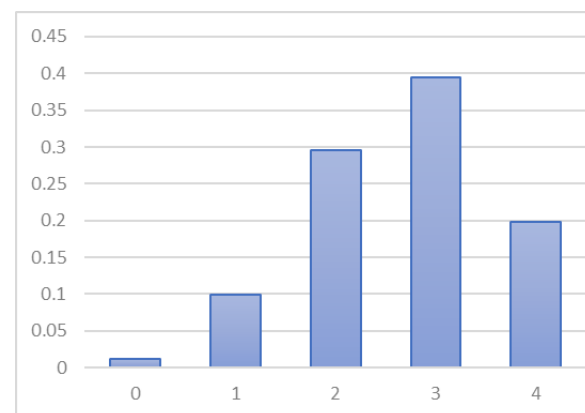
$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - p, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

Binomial Distribution

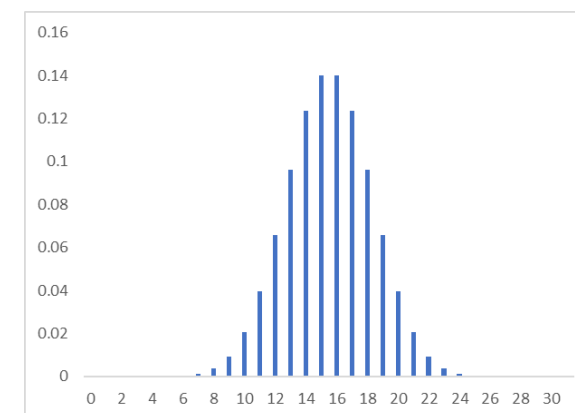
► No. of successes in ' n ' independent and identical Bernoulli trials

Parameters	$n \in \mathbb{N}, p \in (0,1)$
Support	$x \in \{0, 1, 2, \dots, n\}$
PMF	$\binom{n}{x} p^x (1-p)^{n-x}$
CDF	$\sum_{x=0}^x \binom{n}{x} p^x (1-p)^{n-x}$
Mean	np
Variance	$np(1-p)$

pmf ($n = 4, p = 0.6$)



pmf ($n = 30, p = 0.5$)



Questions



Exercise 1 – A student has 60% chances of answering a question correctly. What is the probability that

- a) The student answers exactly 2 out of 4 questions correctly
- b) The student answers at least 2 out of 4 questions correctly

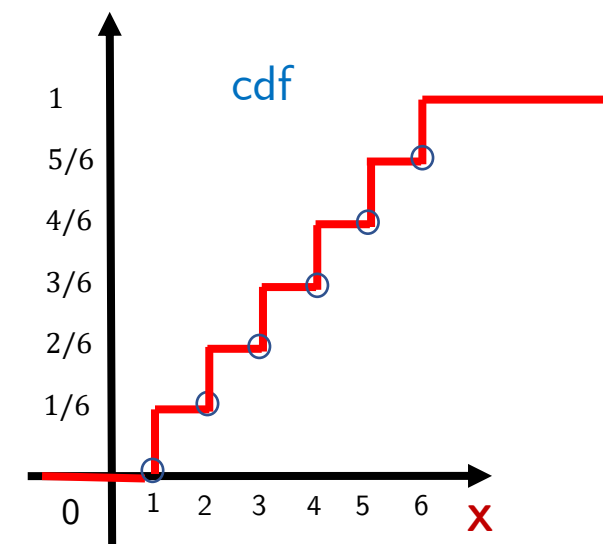
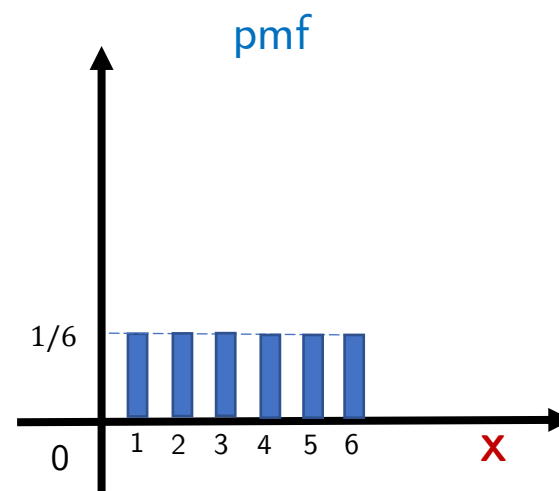
Exercise 2 – A bank has a homogeneous loan portfolio of 5 loans. Each loan amount is \$ 100. Each borrower has a probability of default (PD) = 5%. Assuming independence among default events and no recovery, find

- a) Expected Loss
- b) A conservative loss figure that the bank will exceed only 5% of the time (=won't exceed 95% of the time)

Uniform Distribution

- Used to model a phenomenon with equally likely outcomes : (rolling a die)

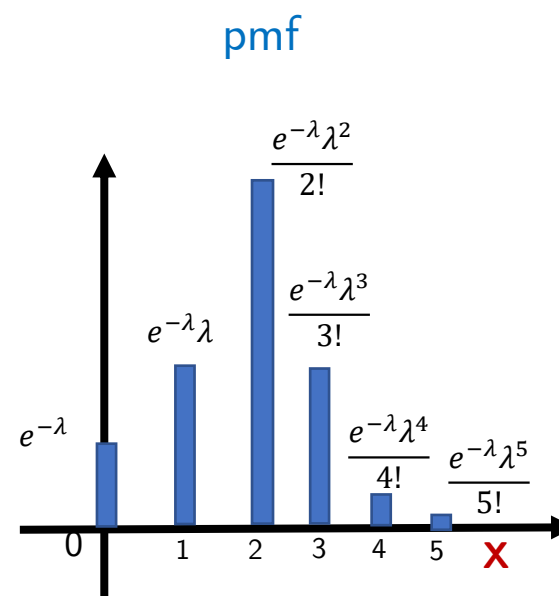
Parameters	a, b
Support	$x \in \{a, a + 1, \dots, b - 1, b\}$ $n = b - a + 1$
PMF	$1/n$
CDF	$\frac{[x] - a + 1}{n}$
Mean	$\frac{a + b}{2}$
Variance	$\frac{n^2 - 1}{12}$



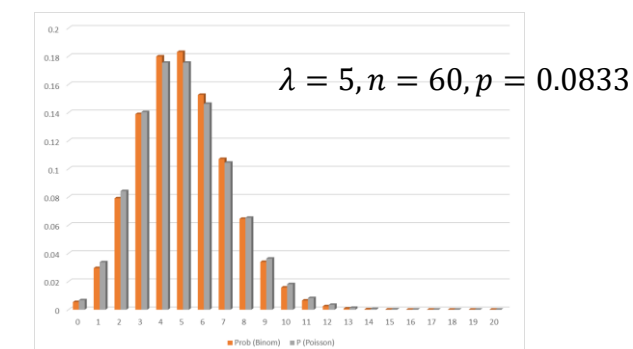
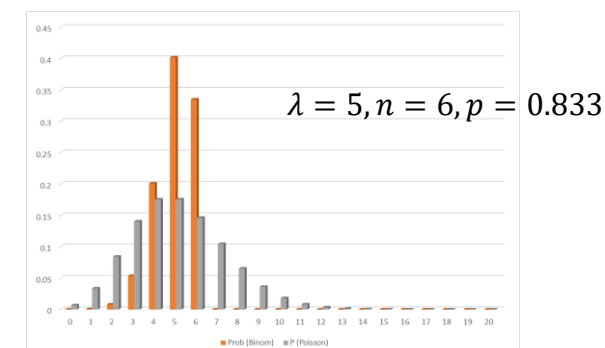
Poisson Distribution

- Used to model number of occurrences of a particular event per unit
 - Customers arriving to a mall on a day ($\lambda = 100$ customers per hr.)
 - Number of defaults in a portfolio of 1 million obligors

Parameters	$\lambda \in (0, \infty)$
Support	$x \in \mathbb{N} \cup \{0\}$
PMF	$\frac{e^{-\lambda} \lambda^x}{x!}$
Mean	λ
Variance	λ



(with $n \rightarrow \infty, p \rightarrow 0$ and $np \rightarrow \lambda$,
Poisson and Binomial converge)



Questions



Exercise 3 – Show that the Poisson pmf is a proper pmf

Exercise 4 – An insurance sales-person sells 3 life insurance policies per week on an average. Use Poisson's law to calculate the probability that in a given month (4 weeks) he will sell

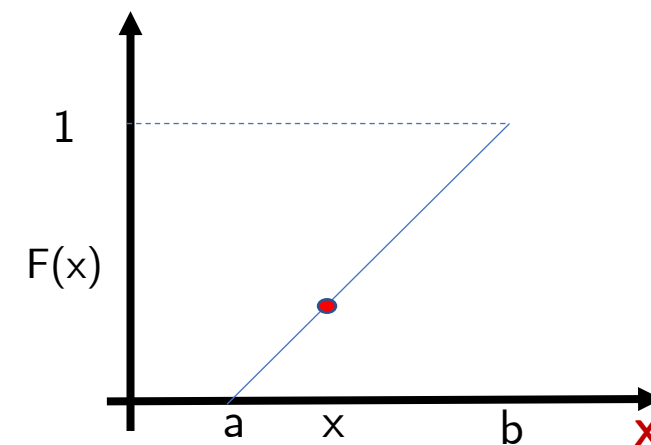
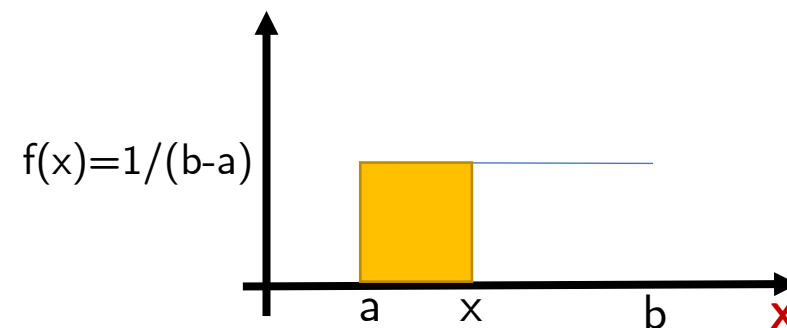
- a. Some policies
- b. 2 or more policies but less than 5 policies.
- c. Assuming that there are 5 working days per week, what is the probability that in a given day he will sell one policy?

(continuous distributions)

Uniform Distribution

- ▶ Mostly used to draw random numbers uniformly between an interval $[0,1]$ for simulation purpose
- ▶ A uniform distribution has bounds $[a, b]$, but a standard uniform has bounds $[0,1]$

Parameters	$-\infty < a < b < \infty$
Support	$x \in [a, b]$
PDF	$\begin{cases} \frac{1}{b-a}, & x \in [a, b] \\ 0, & \text{otherwise} \end{cases}$
CDF	$\begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & x \in [a, b] \\ 1, & x > b \end{cases}$
Mean	$\frac{a+b}{2}$
Variance	$\frac{(b-a)^2}{12}$



Questions



Exercise 5 – For a random variable $X \sim U(0,23)$, find

a) $\mathbb{P}(2 < X < 18)$

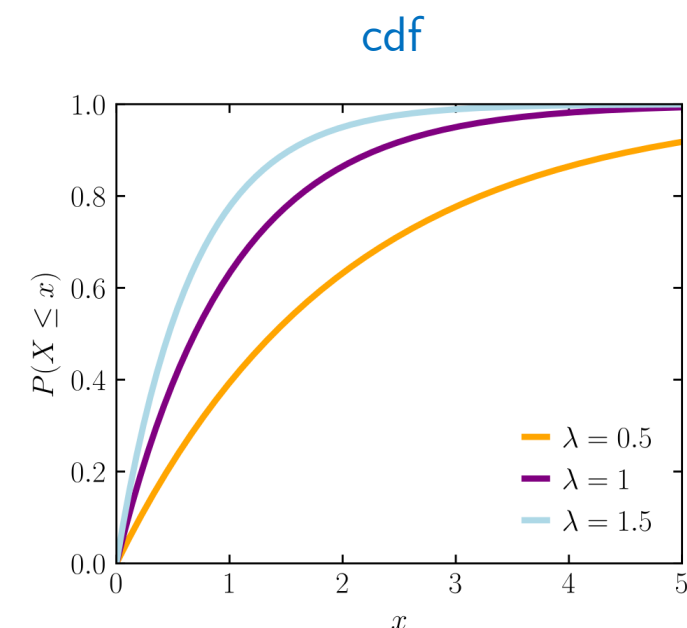
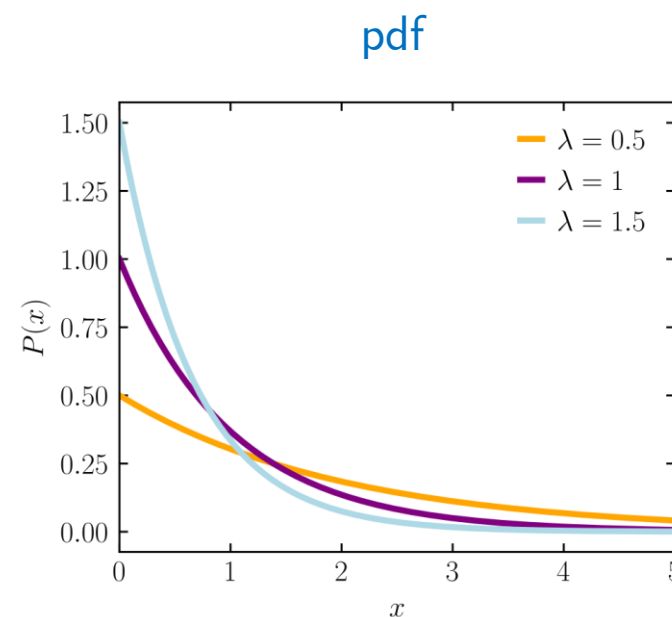
b) $\mathbb{P}(X > 12 \mid X > 8)$

Exercise 6 – If U is standard uniform, prove that $aU + b$ is also uniform $(a, a + b)$. Using this property, what would be the distribution of $2U - 1$?

Exponential Distribution

- ▶ Used to model time to failure/default. It's the basis of survival analysis
- ▶ If arrival is $\text{Poisson}(\lambda)$, then the time elapsed between successive arrival is $\text{Exponential}(\lambda)$

Parameters	$\lambda > 0$
Support	$x \in [0, \infty)$
PDF	$\lambda e^{-\lambda x}$
CDF	$1 - e^{-\lambda x}$
Mean	$1/\lambda$
Variance	$1/\lambda^2$



Questions



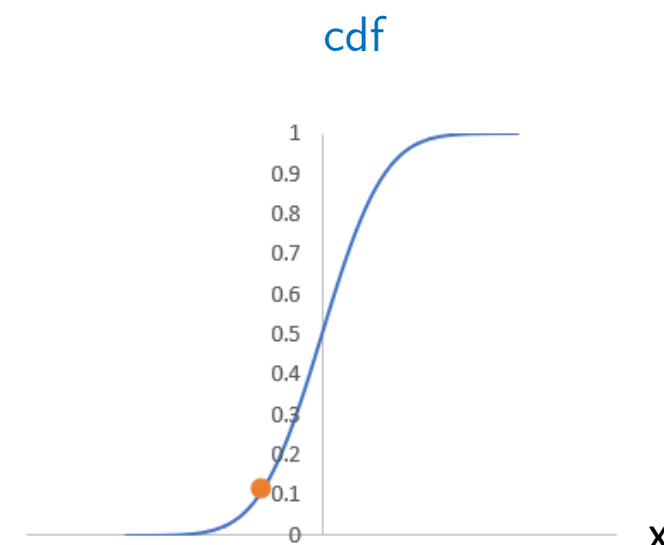
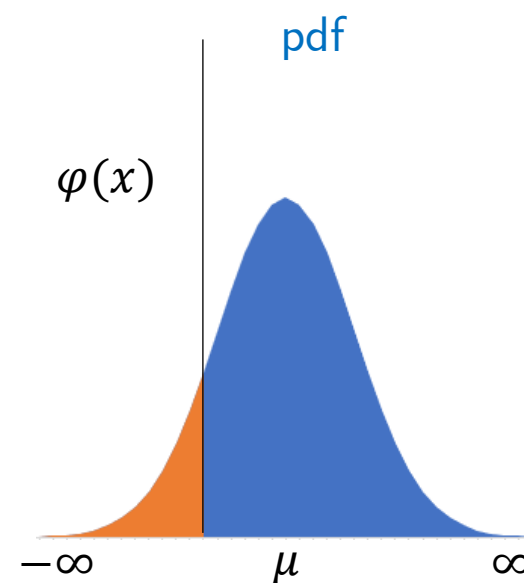
Exercise 6 – Let the hazard rate for a corporate bond be $\lambda = 2\%$ per year. Find the following

- a) Probability of survival till 2 years
- b) Probability of default before 5 years
- c) Probability of default within 2 years to 5 years
- d) Probability of surviving an additional 2 years after surviving for 3 years

Normal Distribution

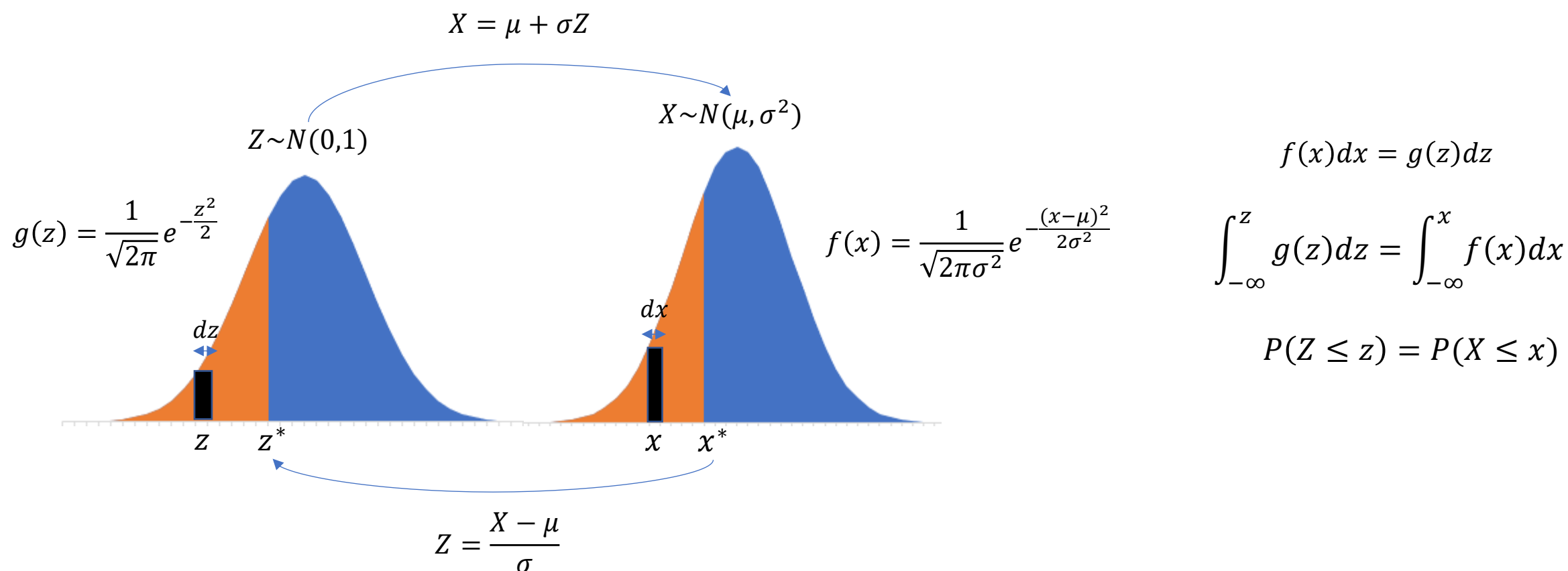
- ▶ The mother of all distributions. We model asset returns as normal.
- ▶ Normal distribution is important because of CLT

Parameters	μ, σ^2
Support	$x \in \mathbb{R}$
PDF	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
CDF	(no closed form formula)
Mean	μ
Variance	σ^2



Normal Distribution vs Standard Normal

- ▶ All probability calculations are done under standard normal distribution
- ▶ The densities are proportional at a given point, however the interval probabilities are the same

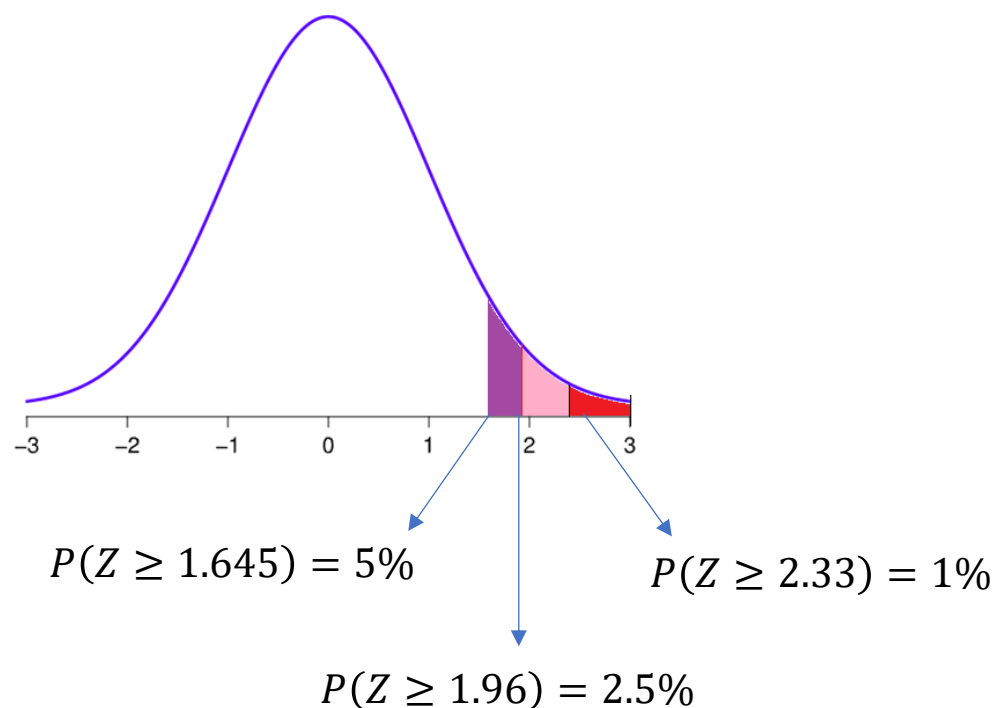


Questions



Exercise 7 – The critical values of the standard normal distribution are given as follows
 $P(Z \geq 1.645) = 5\%$, $P(Z \geq 1.96) = 2.5\%$ and $P(Z \geq 2.33) = 1\%$. Find the following probabilities

- a) $P(Z \geq -1.645)$
- b) $P(-1.96 \leq Z \leq 1.645)$



Questions



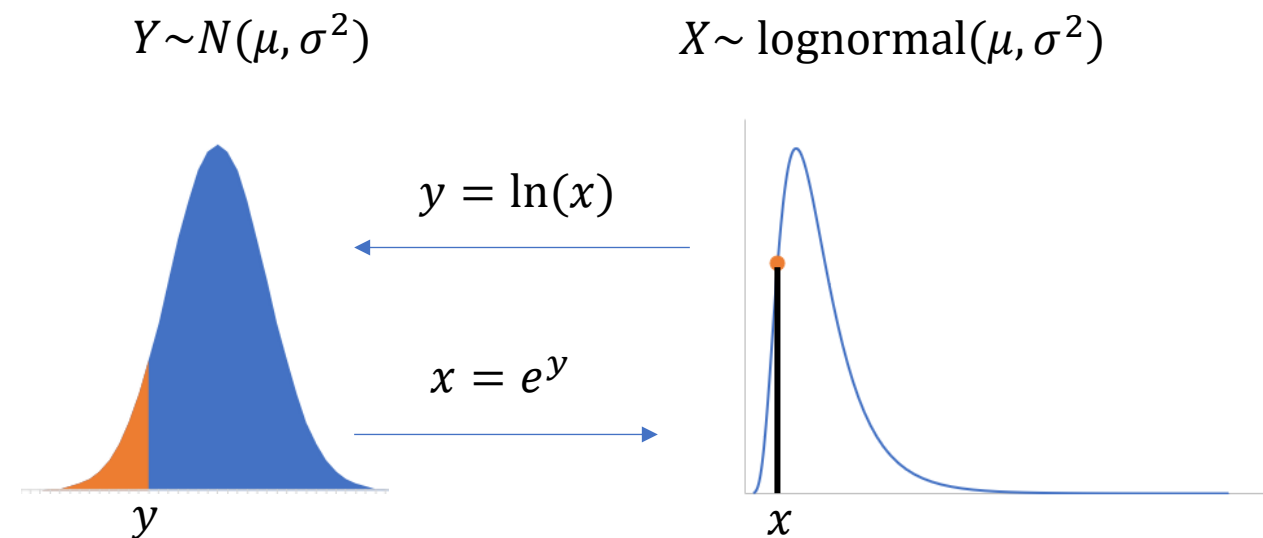
Exercise 8 – Let Z_1 and Z_2 be 2 standard normal random variables. Given the property that the linear combination of 2 normal randoms is a normal random, find $\mathbb{P}(Z_1 > Z_2)$. Does the answer depend on the correlation of Z_1 and Z_2 ?

Exercise 9 – If Z is a standard normal, find $\mathbb{E}[\max(Z, 0)]$ and $\mathbb{E}[Z \mid Z > 0]$

Lognormal Distribution

- ▶ If $Y \sim N(\mu, \sigma^2)$, then $X = e^Y \sim \text{lognorm}(\mu, \sigma^2)$
- ▶ Similarly, if X is lognormal then $\ln X$ is normal
- ▶ We use lognormal distribution to model asset prices

Parameters	μ, σ^2
Support	$x \in \mathbb{R}$
PDF	$\frac{1}{x\sqrt{2\pi\sigma^2}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$
CDF	(no closed form formula)
Mean	$e^{\mu + \frac{1}{2}\sigma^2}$
Variance	$(e^{\sigma^2} - 1)e^{2\mu + \sigma^2}$



Questions



Exercise 10 – Define daily log return of a stock price as

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right)$$

If $r_t \sim N(0.1\%, 5\%)$, and current stock price = \$ 100, what is the expected stock price tomorrow ?